AQA A-Level Further Mathematics (7367) — Topic 1.8

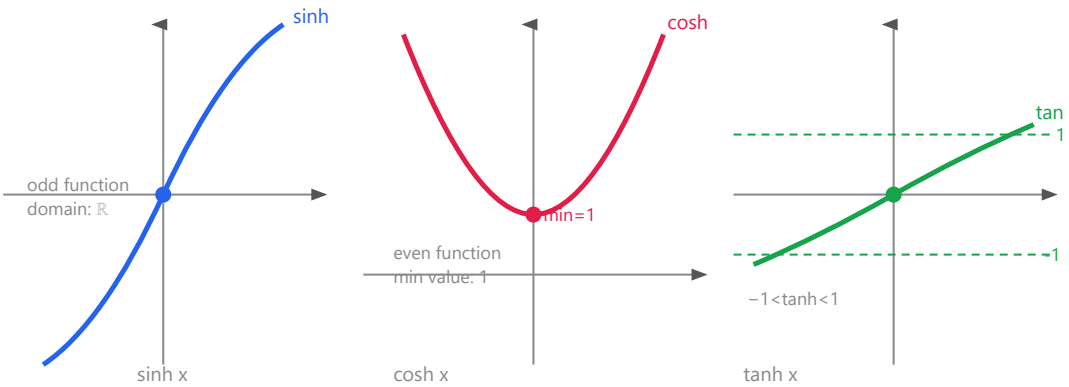
1.8: Hyperbolic Functions

1. Definitions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

The reciprocal functions: $\operatorname{cosech} x = \dfrac{1}{\sinh x}$, $\operatorname{sech} x = \dfrac{1}{\cosh x}$, $\coth x = \dfrac{1}{\tanh x}$

2. Graphs of Hyperbolic Functions



Key Properties Summary

Function	Domain	Range	Parity	Key values
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$	Odd	$\sinh 0 = 0$

$\cosh x$	$\mathbb{R}$	$[1, \infty)$	Even	$\cosh 0 = 1$, min at $x = 0$
$\tanh x$	$\mathbb{R}$	$(-1, 1)$	Odd	$\tanh 0 = 0$; asymptotes $y = \pm 1$

3. Hyperbolic Identities

Fundamental identity: $\cosh^2 x - \sinh^2 x = 1$

Proof: $\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 = \frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{4} = \frac{4}{4} = 1$

Further Identities

Hyperbolic identity	Compare with trig identity
$\cosh^2 x - \sinh^2 x = 1$	$\cos^2 x + \sin^2 x = 1$
$\sinh(2x) = 2 \sinh x \cosh x$	$\sin(2x) = 2 \sin x \cos x$
$\cosh(2x) = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x$	$\cos(2x) = \cos^2 x - \sin^2 x$
$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$	$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$	$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
$1 - \tanh^2 x = \operatorname{sech}^2 x$	$1 + \tan^2 x = \sec^2 x$

Osborne's Rule

Osborne's rule: To convert a trigonometric identity to the corresponding hyperbolic identity: Replace $\cos$ with $\cosh$, replace $\sin$ with $\sinh$, **but replace every product** $\sin^2$ (or any product of two sines) with $-\sinh^2$ (negate).

Note the sign change: $\cos^2\theta + \sin^2\theta = 1$ becomes $\cosh^2 x - \sinh^2 x = 1$ (the product $\sin \cdot \sin$ picks up a minus sign).

AQA Exam Tip: Osborne's rule is a memory aid, not a proof. AQA may ask you to "prove" an identity — use the exponential definitions directly rather than citing Osborne's rule.

4. Inverse Hyperbolic Functions

Logarithmic forms:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2+1}\right), \quad x \in \mathbb{R}$$

$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2-1}\right), \quad x \geq 1$$

$$\operatorname{arctanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1$$

Derivation of $\operatorname{arcsinh} x$**:** Let $y = \sinh^{-1} x$, so $\sinh y = x$, i.e. $\dfrac{e^y - e^{-y}}{2} = x$.

$e^{2y} - 2xe^y - 1 = 0$. Quadratic in e^y:
$e^y = \dfrac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$

Since $e^y > 0$ for all y, take the positive root: $e^y = x + \sqrt{x^2+1}$
$y = \ln(x + \sqrt{x^2+1})$. ✓

5. Derivatives

Derivatives of hyperbolic functions:

$$\frac{d}{dx}(\sinh x) = \cosh x, \qquad \frac{d}{dx}(\cosh x) = \sinh x, \qquad \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

Derivatives of inverse hyperbolic functions:

$$\frac{d}{dx}(\operatorname{arcsinh} x) = \frac{1}{\sqrt{x^2+1}}, \qquad \frac{d}{dx}(\operatorname{arccosh} x) = \frac{1}{\sqrt{x^2-1}}, \qquad \frac{d}{dx}(\operatorname{arctanh} x) = \frac{1}{1-x^2}$$

AQA Exam Tip: Note $\frac{d}{dx}(\cosh x) = \sinh x$ (positive), unlike $\frac{d}{dx}(\cos x) = -\sin x$. This difference catches many students out.

6. Integration Using Hyperbolic Functions

Standard integrals (inverse hyperbolic):

$$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \operatorname{arcsinh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2+a^2}\right) + C$$

$$\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \operatorname{arccosh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2-a^2}\right) + C, \quad x > a$$

$$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{a}\text{arctanh}\!\left(\frac{x}{a}\right) + C = \frac{1}{2a}\ln\!\left(\frac{a+x}{a-x}\right)+C, \quad |x|$$

Example: Find $\int \frac{1}{\sqrt{x^2+4}}\,dx$.

Using formula with $a = 2$: $\operatorname{arcsinh}\left(\frac{x}{2}\right) + C = \ln\left(x + \sqrt{x^2+4}\right) + C$

Example: Find $\int \sinh^2 x\,dx$.

Use identity: $\sinh^2 x = \dfrac{\cosh 2x - 1}{2}$

$\int \dfrac{\cosh 2x - 1}{2}\, dx = \dfrac{\sinh 2x}{4} - \dfrac{x}{2} + C$

Hyperbolic Substitutions

For integrals involving $\sqrt{x^2 + a^2}$, use the substitution $x = a \sinh t$ (since $a^2 \sinh^2 t + a^2 = a^2 \cosh^2 t$).

For integrals involving $\sqrt{x^2 - a^2}$, use $x = a \cosh t$.

Example: Find $\displaystyle\int \sqrt{x^2 + 1}\, dx$ using $x = \sinh t$.

$dx = \cosh t\, dt$, $\sqrt{x^2 + 1} = \sqrt{\sinh^2 t + 1} = \cosh t$

$\int \cosh^2 t\, dt = \int \dfrac{\cosh 2t + 1}{2}\, dt = \dfrac{\sinh 2t}{4} + \dfrac{t}{2} + C$

$= \dfrac{\sinh t \cosh t}{2} + \dfrac{t}{2} + C = \dfrac{x\sqrt{x^2 + 1}}{2} + \dfrac{1}{2}\operatorname{arcsinh} x + C$

7. Solving Hyperbolic Equations

Example: Solve $5 \cosh x - 3 \sinh x = 7$.

Substitute definitions: $5 \cdot \dfrac{e^x + e^{-x}}{2} - 3 \cdot \dfrac{e^x - e^{-x}}{2} = 7$

$\dfrac{5e^x + 5e^{-x} - 3e^x + 3e^{-x}}{2} = 7$

$2e^x + 8e^{-x} = 14$

$e^{2x} - 7e^x + 4 = 0$ (multiply by $e^x/2$)

$e^x = \dfrac{7 \pm \sqrt{49 - 16}}{2} = \dfrac{7 \pm \sqrt{33}}{2}$

$x = \ln\left(\dfrac{7 + \sqrt{33}}{2}\right)$ or $x = \ln\left(\dfrac{7 - \sqrt{33}}{2}\right)$

8. Summary of Key Results

Result	Formula
Fundamental identity	$\cosh^2 x - \sinh^2 x = 1$
$\operatorname{arcsinh} x$	$\ln(x + \sqrt{x^2+1})$
$\operatorname{arccosh} x$	$\ln(x + \sqrt{x^2-1})$, $x \geq 1$
$\operatorname{arctanh} x$	$\frac{1}{2}\ln\frac{1+x}{1-x}$, $\lvert x\rvert < 1$
$\frac{d}{dx}\sinh x$	$\cosh x$
$\frac{d}{dx}\cosh x$	$\sinh x$
$\frac{d}{dx}\tanh x$	$\operatorname{sech}^2 x$
$\int 1/\sqrt{x^2+a^2}\,dx$	$\operatorname{arcsinh}(x/a) + C$
$\int 1/\sqrt{x^2-a^2}\,dx$	$\operatorname{arccosh}(x/a) + C$

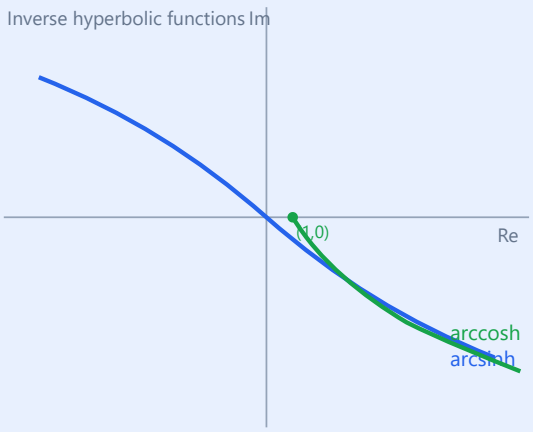
9. Practice Checklist

- Can you define $\sinh$, $\cosh$, $\tanh$ in terms of exponentials?
- Can you prove the identity $\cosh^2 x - \sinh^2 x = 1$ from the definitions?
- Can you state and use the double-angle hyperbolic identities?
- Can you derive the logarithmic forms of the inverse functions?
- Can you differentiate and integrate hyperbolic and inverse hyperbolic functions?
- Can you solve equations like $a\cosh x + b\sinh x = c$ by substituting exponential definitions?

Inverse Hyperbolic Functions — Graphs and Domains

Function	Domain	Range	Logarithmic form

$\operatorname{arcsinh} x$	$\mathbb{R}$	$\mathbb{R}$	$\ln(x+\sqrt{x^2+1})$
$\operatorname{arccosh} x$	$x \geq 1$	$y \geq 0$	$\ln(x+\sqrt{x^2-1})$
$\operatorname{arctanh} x$	$\lvert x\rvert < 1$	$\mathbb{R}$	$\frac{1}{2}\ln\frac{1+x}{1-x}$



Derivatives of inverse hyperbolics:

$$\frac{d}{dx}\operatorname{arcsinh} x = \frac{1}{\sqrt{x^2+1}}, \qquad \frac{d}{dx}\operatorname{arccosh} x = \frac{1}{\sqrt{x^2-1}}, \qquad \frac{d}{dx}\operatorname{arctanh} x = \frac{1}{1-x^2}$$

Integration Using Hyperbolic Functions

Hyperbolic substitutions and direct standard forms:

Example 1: $\int \frac{1}{\sqrt{x^2+4}}\,dx$. Let $x = 2\sinh t$, $dx = 2\cosh t\,dt$, $\sqrt{x^2+4} = 2\cosh t$:

$$\int \frac{2\cosh t}{2\cosh t}\,dt = t + C = \operatorname{arcsinh}(x/2) + C = \ln\left(\frac{x}{2} + \sqrt{\frac{x^2}{4}+1}\right) + C$$

Example 2: $\int \frac{1}{4-x^2}\,dx$ for $\lvert x\rvert < 2$. Let $x = 2\tanh t$, $dx = 2\operatorname{sech}^2 t\,dt$, $4 - x^2 = 4\operatorname{sech}^2 t$:

$$\int \frac{2\text{sech}^2 t}{4\text{sech}^2 t}\,dt = \frac{t}{2} + C = \frac{1}{2}\text{arctanh}(x/2) + C = \frac{1}{4}\ln\frac{2+x}{2-x} + C$$

Example 3: $\int_0^1 \sqrt{1+x^2}\,dx$. Let $x = \sinh t$: $dx = \cosh t\,dt$, limits 0 to $\text{arcsinh}(1) = \ln(1+\sqrt{2})$:

$$\int_0^{\ln(1+\sqrt{2})} \cosh^2 t\,dt = \left[\frac{\sinh 2t}{4} + \frac{t}{2}\right]_0^{\ln(1+\sqrt{2})} = \frac{\sqrt{2}}{2} + \frac{\ln(1+\sqrt{2})}{2}$$

The Catenary — Application of $\cosh$

A uniform flexible chain hanging under gravity takes the shape of a **catenary**:

$$y = a\cosh\left(\frac{x}{a}\right)$$

where a is the height of the lowest point above the x-axis direction. Properties:

- Minimum at $(0, a)$ with $y'(0) = 0$.
- $y'' = \frac{1}{a}\cosh(x/a) = \frac{y}{a^2}$ — curvature is proportional to y.
- Arc length from $-L$ to L: $2a\sinh(L/a)$.
- The Gateway Arch in St Louis is (approximately) an inverted catenary.

Surface area of a catenary: When $y = \cosh x$ is rotated about the x-axis from $-a$ to a:

$$S = 2\pi\int_{-a}^{a}\cosh x\sqrt{1+\sinh^2 x}\,dx = 2\pi\int_{-a}^{a}\cosh^2 x\,dx = \pi\left[\sinh(2a)/2 + 2a\right]$$

(using the double-angle identity $\cosh^2 x = (1+\cosh 2x)/2$).

AQA Exam Technique — Hyperbolic Functions

Key tips:

- **Identity proofs:** Start from exponential definitions. Don't try to prove by analogy with trig — show the working explicitly.
- **Osborne's rule:** Replace $\cos \to \cosh$, $\sin \to \sinh$, but replace $\sin^2 \to -\sinh^2$ (change sign when $\sin^2$ appears). This gives the correct hyperbolic identities.
- **Inverse forms:** For the logarithmic derivation, set $y = \sinh^{-1} x$ so $\sinh y = x$, write as exponential, solve the quadratic in e^y, discard the negative root.
- **Integration:** Recognise $\dfrac{1}{\sqrt{a^2+x^2}} \to \operatorname{arcsinh}(x/a)$ and $\dfrac{1}{\sqrt{x^2-a^2}} \to \operatorname{arccosh}(x/a)$ as standard forms.
- **Equations $a\cosh x + b\sinh x = c$:** Substitute exponential forms and rearrange to a quadratic in e^x. Discard solutions where $e^x \leq 0$.

Standard integral	Result
$\int \sinh x \, dx$	$\cosh x + C$
$\int \cosh x \, dx$	$\sinh x + C$
$\int \tanh x \, dx$	$\ln(\cosh x) + C$
$\int \dfrac{1}{\sqrt{x^2+a^2}} \, dx$	$\operatorname{arcsinh}(x/a) + C$
$\int \dfrac{1}{\sqrt{x^2-a^2}} \, dx$	$\operatorname{arccosh}(x/a) + C$
$\int \dfrac{1}{a^2-x^2} \, dx$	$\dfrac{1}{a}\operatorname{arctanh}(x/a) + C = \dfrac{1}{2a}\ln\dfrac{a+x}{a-x} + C$

Hyperbolic Identities — Proofs

All hyperbolic identities follow from the definitions $\cosh x = \dfrac{e^x + e^{-x}}{2}$, $\sinh x = \dfrac{e^x - e^{-x}}{2}$.

Proof of $\cosh^2 x - \sinh^2 x = 1$**:**

$\cosh^2 x = \dfrac{e^{2x} + 2 + e^{-2x}}{4}, \quad \sinh^2 x = \dfrac{e^{2x} - 2 + e^{-2x}}{4}.$

$\cosh^2 x - \sinh^2 x = \dfrac{4}{4} = 1.$ ✓

Proof of $\sinh(A+B) = \sinh A \cosh B + \cosh A \sinh B$**:**

$\sinh A \cosh B = \dfrac{e^A - e^{-A}}{2} \cdot \dfrac{e^B + e^{-B}}{2} = \dfrac{e^{A+B} + e^{A-B} - e^{-A+B} - e^{-(A+B)}}{4}$

$\cosh A \sinh B = \dfrac{e^{A+B} - e^{A-B} + e^{-A+B} - e^{-(A+B)}}{4}$

Adding: $\dfrac{2e^{A+B} - 2e^{-(A+B)}}{4} = \dfrac{e^{A+B} - e^{-(A+B)}}{2} = \sinh(A+B).$ ✓

Proof of double angle formula $\cosh 2x = 2\cosh^2 x - 1$**:**

$\cosh 2x = \cosh(x+x) = \cosh^2 x + \sinh^2 x = \cosh^2 x + \cosh^2 x - 1 = 2\cosh^2 x - 1.$ ✓

Solving Hyperbolic Equations

Example 1 — Solve $\cosh x = 2$**:**

$\dfrac{e^x + e^{-x}}{2} = 2 \Rightarrow e^x + e^{-x} = 4$. Let $u = e^x > 0$: $u + \frac{1}{u} = 4 \Rightarrow u^2 - 4u + 1 = 0$.

$u = 2 \pm \sqrt{3}$. Both positive: $x = \ln(2+\sqrt{3})$ or $x = \ln(2-\sqrt{3}) = -\ln(2+\sqrt{3})$.

So $x = \pm\ln(2+\sqrt{3}) = \pm\operatorname{arccosh} 2.$ ✓

Example 2 — Solve $3\sinh x - \cosh x = 1$**:**

Write in terms of exponentials: $\dfrac{3(e^x - e^{-x})}{2} - \dfrac{e^x + e^{-x}}{2} = 1.$

$\dfrac{3e^x - 3e^{-x} - e^x - e^{-x}}{2} = 1 \Rightarrow 2e^x - 4e^{-x} = 2 \Rightarrow e^x - 2e^{-x} = 1.$

Let $u = e^x$: $u - \frac{2}{u} = 1 \Rightarrow u^2 - u - 2 = 0 \Rightarrow (u-2)(u+1) = 0$.
$u = 2$ (since $u > 0$), so $x = \ln 2$.

Example 3 — Solve $\tanh x = \frac{1}{2}$:
$\operatorname{arctanh}\left(\frac{1}{2}\right) = \dfrac{1}{2}\ln\dfrac{1+\frac{1}{2}}{1-\frac{1}{2}} = \dfrac{1}{2}\ln 3.$

Inverse Hyperbolic Functions — Log Forms and Proofs

Function	Domain	Log form
$\operatorname{arcsinh} x$	$x \in \mathbb{R}$	$\ln\left(x + \sqrt{x^2+1}\right)$
$\operatorname{arccosh} x$	$x \geq 1$	$\ln\left(x + \sqrt{x^2-1}\right)$
$\operatorname{arctanh} x$	$\lvert x\rvert < 1$	$\dfrac{1}{2}\ln\dfrac{1+x}{1-x}$

Proof that $\operatorname{arcsinh} x = \ln(x + \sqrt{x^2+1})$:
Let $y = \operatorname{arcsinh} x$, so $\sinh y = x$, i.e. $\dfrac{e^y - e^{-y}}{2} = x$.
$e^{2y} - 2xe^y - 1 = 0$. Quadratic in e^y: $e^y = \dfrac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$.
Since $e^y > 0$ and $\sqrt{x^2+1} > \lvert x\rvert$, take $+$: $e^y = x + \sqrt{x^2+1}$, so $y = \ln(x + \sqrt{x^2+1})$. ✓

Differentiation of Inverse Hyperbolic Functions

Function	Derivative
$\operatorname{arcsinh} x$	$\dfrac{1}{\sqrt{1+x^2}}$

$\operatorname{arccosh} x$	$\dfrac{1}{\sqrt{x^2-1}} \quad (x>1)$
$\operatorname{arctanh} x$	$\dfrac{1}{1-x^2} \quad (\lvert x\rvert<1)$
$\operatorname{arcsinh}\dfrac{x}{a}$	$\dfrac{1}{\sqrt{a^2+x^2}}$
$\operatorname{arccosh}\dfrac{x}{a}$	$\dfrac{1}{\sqrt{x^2-a^2}} \quad (x>a)$

Proof that $\dfrac{d}{dx}(\operatorname{arcsinh} x) = \dfrac{1}{\sqrt{1+x^2}}$:

Using log form: $\dfrac{d}{dx}\ln(x+\sqrt{x^2+1}) = \dfrac{1+\frac{x}{\sqrt{x^2+1}}}{x+\sqrt{x^2+1}} = \dfrac{\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}}}{x+\sqrt{x^2+1}} = \dfrac{1}{\sqrt{x^2+1}}$. ✓

Hyperbolic Substitution Integrals — Further Examples

Example 1: $\displaystyle\int \sqrt{x^2+4}\,dx$. Let $x = 2\sinh u$, $dx = 2\cosh u\,du$.

$\sqrt{x^2+4} = \sqrt{4\sinh^2 u+4} = 2\cosh u$.

$$\int 2\cosh u \cdot 2\cosh u\,du = 4\int \cosh^2 u\,du = 4\int \frac{1+\cosh 2u}{2}\,du = 2u+\sinh 2u + C$$

$$= 2u + 2\sinh u\cosh u + C = 2\operatorname{arcsinh}\frac{x}{2} + \frac{x\sqrt{x^2+4}}{2} + C$$

Example 2: $\displaystyle\int_2^3 \frac{1}{\sqrt{x^2-1}}\,dx$. Let $x = \cosh u$, $dx = \sinh u\,du$.

$\sqrt{x^2-1} = \sinh u$. So $\displaystyle\int \frac{\sinh u}{\sinh u}\,du = u + C = \operatorname{arccosh} x + C$.

$[\operatorname{arccosh} x]_2^3 = \operatorname{arccosh} 3 - \operatorname{arccosh} 2 = \ln(3+2\sqrt{2}) - \ln(2+\sqrt{3})$

Osborn's Rule

Osborn's Rule: To convert a trigonometric identity to the corresponding hyperbolic identity, replace $\cos\theta$ with $\cosh\theta$, $\sin\theta$ with $i\sinh\theta$ (or equivalently: replace any product of two $\sin$ terms with minus the corresponding $\sinh$ product).
For example: $\cos^2\theta + \sin^2\theta = 1 \to \cosh^2\theta - \sinh^2\theta = 1$.
$\cos^2\theta - \sin^2\theta = \cos 2\theta \to \cosh^2\theta + \sinh^2\theta = \cosh 2\theta$.

Osborn's Rule caveat: It applies to basic identities, not to integrals or derivatives. Don't blindly apply it to results like $\int \sin^2\theta\, d\theta$.

Graphs of Hyperbolic Functions

Key features:

- $\cosh x$: even function; minimum value 1 at $x = 0$; $\cosh x \to \infty$ as $x \to \pm\infty$. Graph is a catenary.
- $\sinh x$: odd function; passes through origin; strictly increasing; $\sinh x \approx x$ near 0.
- $\tanh x$: odd function; range $(-1, 1)$; horizontal asymptotes $y = \pm 1$; passes through origin with slope 1.
- $\operatorname{sech} x = 1/\cosh x$: even; max value 1 at $x = 0$; bell-shaped.
- $\operatorname{arcsinh} x$: odd; defined for all x; looks like $\ln(2x)$ for large x.
- $\operatorname{arccosh} x$: defined for $x \geq 1$; starts at $(1, 0)$.
- $\operatorname{arctanh} x$: odd; defined for $|x| < 1$; vertical asymptotes at $x = \pm 1$.

Applications of Hyperbolic Functions — Mechanics

The Catenary: A flexible chain or cable hanging under gravity takes the shape $y = a\cosh\frac{x}{a}$. The arc length of the catenary from $x = 0$ to $x = L$ is:

$$s = a \sinh \frac{L}{a}$$

The tension at a point is $T = T_0 \cosh \frac{x}{a}$ where T_0 is the tension at the lowest point.

Example — Chain of length 20 m with lowest point 4 m below endpoints:
The chain hangs between poles distance $2d$ apart. With $y = a\cosh(x/a)$:
At $x = d$: $y = a + 4$ (4 m above lowest). Arc length $= 2a\sinh(d/a) = 20$, so $a\sinh(d/a) = 10$
.
Also $a\cosh(d/a) = a + 4$. Dividing: $\tanh(d/a) = 10/(a+4)$. Solve numerically.

Series Expansions of Hyperbolic Functions

Maclaurin series (all valid for all x):

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

Compare with: $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$ and $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$.
The hyperbolic functions have all positive signs — consistent with $\cosh x = \cos(ix)$ and $\sinh x = -i\sin(ix)$.

Example — Approximate $\cosh(0.1)$:
$\cosh(0.1) \approx 1 + \frac{0.01}{2} + \frac{0.0001}{24} = 1 + 0.005 + 0.0000042 \approx 1.0050042$.
(Exact to 7 d.p.: 1.0050042.)

Connection: Trigonometric and Hyperbolic Functions

Using complex numbers ($i = \sqrt{-1}$):

$$\cos(ix) = \cosh x, \qquad \sin(ix) = i \sinh x$$

$$\cosh(ix) = \cos x, \qquad \sinh(ix) = i \sin x$$

This means: all hyperbolic identities follow from trig identities by the substitution $\theta \to ix$.

Example: Derive $\cosh(A+B) = \cosh A \cosh B + \sinh A \sinh B$ from $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$.
Set $\alpha = iA, \beta = iB$: $\cos(i(A+B)) = \cos(iA)\cos(iB) - \sin(iA)\sin(iB)$.
$\cosh(A+B) = \cosh A \cosh B - (i\sinh A)(i \sinh B) = \cosh A \cosh B + \sinh A \sinh B$.
✓

Complete Identities Summary

Identity	Value
$\cosh^2 x - \sinh^2 x$	1
$\text{sech}^2 x + \tanh^2 x$	1
$\coth^2 x - \text{csch}^2 x$	1
$\cosh 2x$	$\cosh^2 x + \sinh^2 x = 2\cosh^2 x - 1 = 1 + 2\sinh^2 x$
$\sinh 2x$	$2 \sinh x \cosh x$
$\tanh 2x$	$\dfrac{2\tanh x}{1+\tanh^2 x}$
$\cosh(A+B)$	$\cosh A \cosh B + \sinh A \sinh B$
$\sinh(A+B)$	$\sinh A \cosh B + \cosh A \sinh B$

$\tanh(A+B)$	$\dfrac{\tanh A + \tanh B}{1 + \tanh A \tanh B}$

Integration by Parts Using Hyperbolic Functions

Example 1: $\int x \sinh x \, dx$
IBP: $u = x,\ dv = \sinh x \, dx \Rightarrow v = \cosh x.$
$= x\cosh x - \int \cosh x \, dx = x\cosh x - \sinh x + C$

Example 2: $\int x \cosh x \, dx$
IBP: $u = x,\ dv = \cosh x \, dx \Rightarrow v = \sinh x.$
$= x\sinh x - \int \sinh x \, dx = x\sinh x - \cosh x + C$

Example 3: $\int \sinh^2 x \, dx$
Use $\sinh^2 x = \dfrac{\cosh 2x - 1}{2}$:
$$\int \sinh^2 x \, dx = \frac{\sinh 2x}{4} - \frac{x}{2} + C = \frac{\sinh x \cosh x}{2} - \frac{x}{2} + C$$

Example 4: $\int \cosh^2 x \, dx$
Use $\cosh^2 x = \dfrac{\cosh 2x + 1}{2}$:
$$\int \cosh^2 x \, dx = \frac{\sinh 2x}{4} + \frac{x}{2} + C = \frac{\sinh x \cosh x}{2} + \frac{x}{2} + C$$

Further Worked Differentiation Examples

Example 1 — Chain rule: $\dfrac{d}{dx}\cosh(3x^2+1)$
$= \sinh(3x^2+1) \cdot 6x = 6x\sinh(3x^2+1)$

Example 2 — Product rule: $\frac{d}{dx}[x^2 \sinh x]$
$= 2x \sinh x + x^2 \cosh x$

Example 3 — Implicit differentiation: Given $\cosh y = x$, find $\frac{dy}{dx}$.
$\sinh y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\sinh y}$.
Since $\cosh y = x$: $\sinh y = \sqrt{x^2 - 1}$ (for $y \geq 0$).
So $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}$ — confirming the derivative of $\operatorname{arccosh} x$.

Hyperbolic Functions — AQA Exam Technique

- **Equations:** If given $\cosh x = k$ with $k > 1$, write $\pm \operatorname{arccosh} k = \pm \ln(k + \sqrt{k^2 - 1})$. Both signs are valid unless domain restricted.
- **Integration:** Recognise $\frac{1}{\sqrt{x^2 \pm a^2}}$ and use arcsinh/arccosh. For $\frac{1}{a^2 - x^2}$, use arctanh (when \ (|x|
- **Completing the square:** If the denominator is $\sqrt{x^2 + 4x + 8} = \sqrt{(x+2)^2 + 4}$, substitute $u = x + 2$ first.
- **Identity derivation:** Always go back to definitions e^x if unsure — expand both sides from scratch.
- **Log form answers:** AQA often accepts both arcsinh/arccosh notation and the logarithmic form. Show they are equivalent for full marks.
- **Osborn's Rule:** Useful for guessing identities, but always verify from definitions for a proof question.

Completing the Square and Hyperbolic Integration

Example 1: $\displaystyle\int \frac{1}{\sqrt{x^2+6x+10}}\,dx$

$x^2+6x+10=(x+3)^2+1$. Let $u=x+3$:

$$\int \frac{1}{\sqrt{u^2+1}}\,du = \operatorname{arcsinh} u + C = \ln\left(x+3+\sqrt{x^2+6x+10}\right)+C$$

Example 2: $\displaystyle\int \frac{1}{\sqrt{x^2-4x+3}}\,dx$

$x^2-4x+3=(x-2)^2-1$. Let $u=x-2$:

$$\int \frac{1}{\sqrt{u^2-1}}\,du = \operatorname{arccosh} u + C = \ln\left(x-2+\sqrt{x^2-4x+3}\right)+C \quad (x>3)$$

Example 3: $\displaystyle\int \frac{1}{3-x^2}\,dx \quad (|x|<\sqrt{3})$

Write as $\dfrac{1}{(\sqrt{3})^2-x^2}$. Use $\operatorname{arctanh}$ formula:

$$= \frac{1}{2\sqrt{3}}\ln\frac{\sqrt{3}+x}{\sqrt{3}-x}+C = \frac{1}{\sqrt{3}}\operatorname{arctanh}\frac{x}{\sqrt{3}}+C$$

Hyperbolic Functions — Numerical Values

Useful values to remember or compute quickly:

- $\cosh 0 = 1,\ \sinh 0 = 0,\ \tanh 0 = 0$
- $\cosh 1 \approx 1.543,\ \sinh 1 \approx 1.175,\ \tanh 1 \approx 0.762$
- $\operatorname{arcsinh} 1 = \ln(1+\sqrt{2}) \approx 0.881$
- $\operatorname{arccosh} 2 = \ln(2+\sqrt{3}) \approx 1.317$
- $\operatorname{arctanh}\frac{1}{2} = \frac{1}{2}\ln 3 \approx 0.549$
- As $x\to\infty$: $\cosh x \approx \sinh x \approx \dfrac{e^x}{2}$, \;\tanh x\to 1\).
- The parametric equations $x=\cosh t,\ y=\sinh t$ satisfy $x^2-y^2=1$, tracing the right branch of the unit hyperbola — this is the geometric origin of the name "hyperbolic functions".
- Compare with circular functions: $x=\cos t,\ y=\sin t$ satisfies $x^2+y^2=1$ (unit circle). The parameter t in the hyperbolic case represents twice the area swept out, not an angle.

- For AQA, you must know: all basic derivatives and integrals of $\sinh$, $\cosh$, $\tanh$ and their inverses; how to prove identities from e^x definitions; and how to solve equations using the quadratic-in-e^x technique.
- The identity $\cosh^2 x - \sinh^2 x = 1$ is the hyperbolic analogue of $\cos^2\theta + \sin^2\theta = 1$. Use it whenever you need to simplify $\cosh^2$ or $\sinh^2$ — analogous to how $\sin^2 + \cos^2 = 1$ is used in trigonometry.